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Experiment 6

Aim: To study Agile Methodology and Test case Management using JIRA tool.

Theory:

1. Agile Methodology

Agile is an iterative and incremental approach to software development and project management. Unlike traditional sequential models (like the Waterfall model), Agile focuses on flexibility, continuous improvement, and rapid delivery of functional software.

Core Principles of Agile:

- Iterative Development: Work is divided into smaller, manageable chunks known as iterations or sprints, typically lasting 1 to 4 weeks.
- Customer Collaboration: Continuous feedback from stakeholders ensures the product meets user needs.
- Adaptability: Agile teams embrace changing requirements, even late in the development process.
- Cross-functional Teams: Developers, testers, and designers work collaboratively rather than in isolated silos.

2. Introduction to Jira

Jira, developed by Atlassian, is one of the most widely used project management and issue-tracking tools designed specifically to support Agile methodologies. It allows teams to plan, track, and release software efficiently. Jira supports various Agile frameworks, primarily Scrum and Kanban.

Key Components in Jira:

- Project: A collection of issues held in a common workspace. As seen in your experiment, the project is tracked using a Kanban board.
- Issue: The basic unit of work in Jira. Everything tracked in Jira is an issue, which can represent a bug, a project task, or a helpdesk ticket.
- Issue Types:
    - Epic: A large body of work that can be broken down into smaller tasks.
    - Story / Task: A single, manageable piece of work assigned to a specific team member (e.g., "Write factorial code" or "Make a new code").
    - Sub-task: A smaller, granular task that is part of a larger Task or Epic (e.g., "Finalize qa testing" falling under "Do qa testing").
- Workflow: Represents the sequential path an issue takes from creation to completion. A standard workflow transitions an issue's status from TO DO, to IN PROGRESS, and finally to DONE.

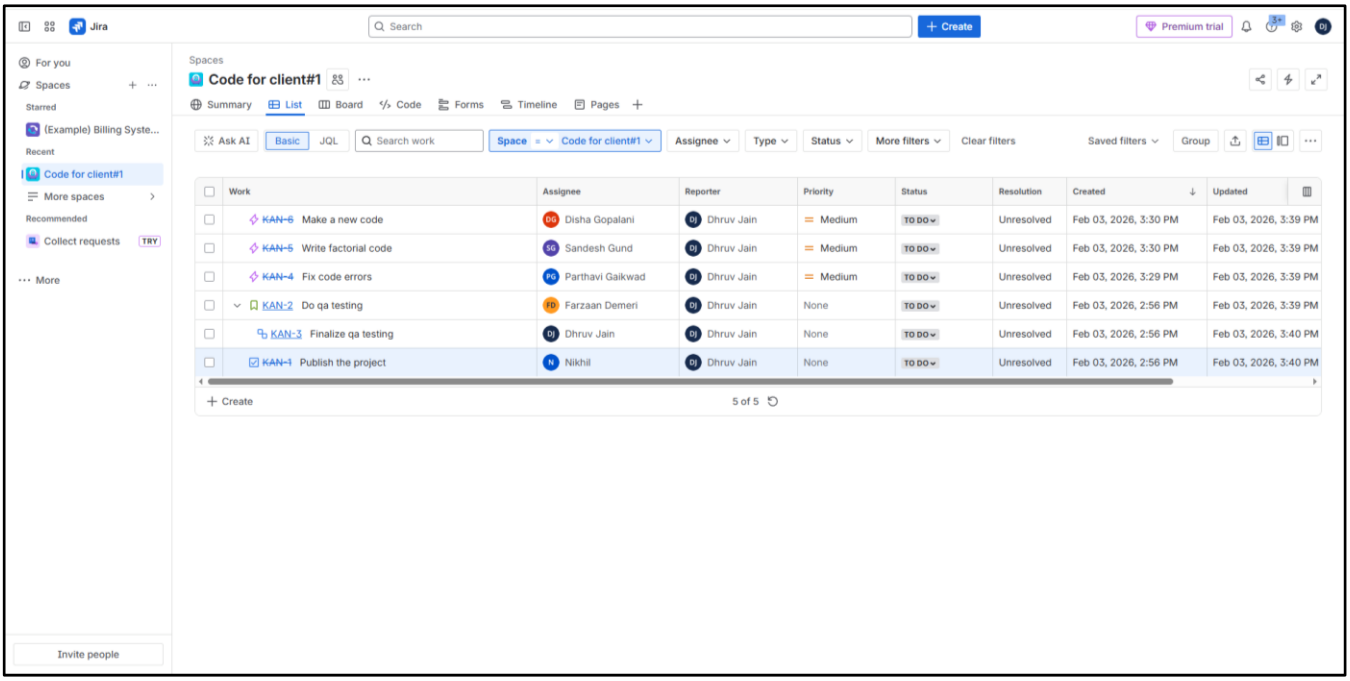
3. Test Case Management in Jira

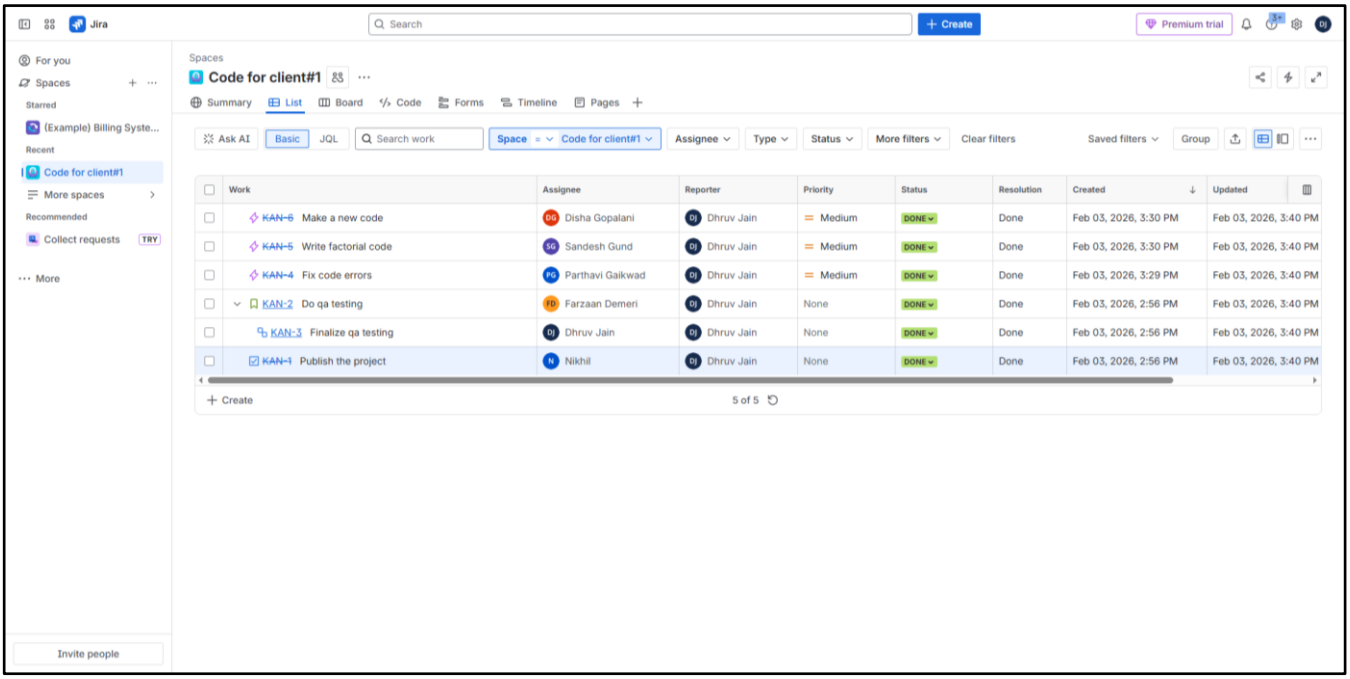
Test case management involves planning, authoring, executing, and tracking software tests. While Jira is fundamentally an issue tracker, it is heavily used for test management in Agile environments.

How Jira is used for QA and Testing:

- Bug Tracking: QA engineers log defects as specific "Bug" issue types (e.g., "Fix code errors"). These bugs are assigned priorities and tracked until resolved.
- Traceability: Bugs and testing tasks can be directly linked to the original developer tasks or Epics, ensuring complete visibility of what is causing an issue.

Implementation:





Conclusion:

Hence, we have successfully studied Agile Methodology and Test case Management using JIRA tool.

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